Residual collector with distinguished points and an equivariant walk (priced)

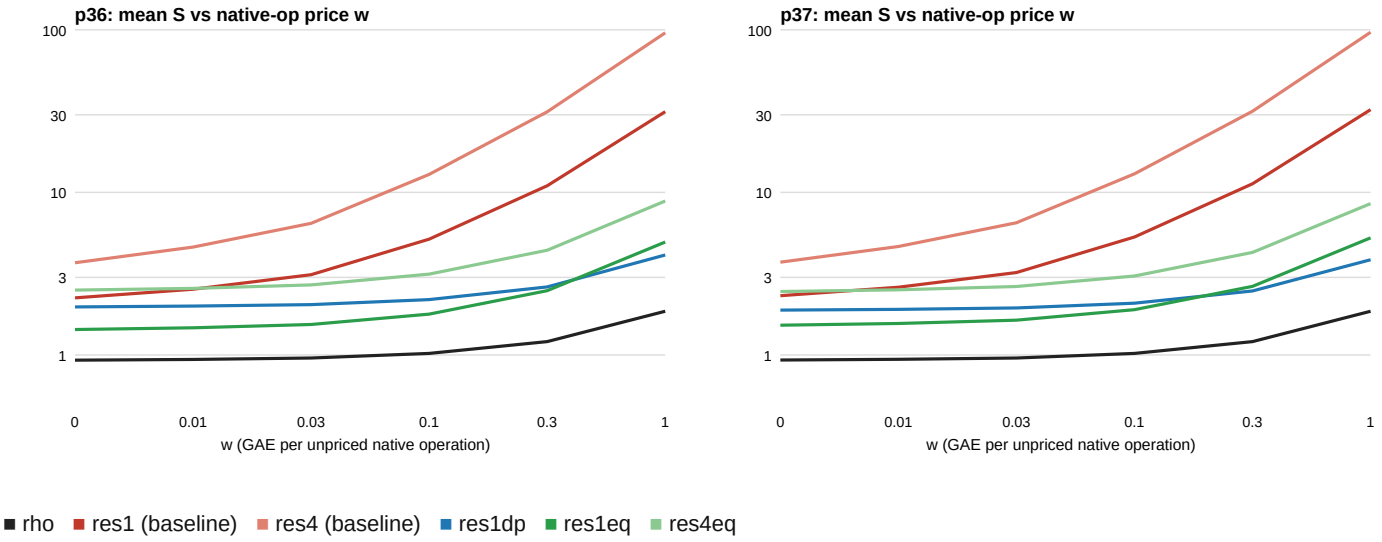
Correction. `research/ic_residual_dp_20261101` left the per-step distinguished-point test out of the priced counters, so its w > 0 figures for `residual-fused-dp` understate priced work. This study reruns with the per-step work priced (`*_uncharged`, 1 GAE per operation at w = 1, like rho's canonicalisation) and supersedes those figures.

What was implemented (`src/cryptanalysis/ic_framework/residual.rs`; baseline `targets=residual` unchanged): `residual-fused-dp` (one addition per step from a $J_j - P_i$ table, distinguished points with $b = \lfloor \log_2 r / 2 \rfloor - 9 = 8$ on p36/p37) and `residual-fused-eq` (a negation-equivariant walk, so L and −L paths merge, on 2016 pair sums $J_u + J_v - P_i$ with a 2-cycle escape and distinguished points). Priced per step: DP test (both), orientation read and one negation per flipped step (eq). Setup additions and the extra escape additions are counted group operations.

Counted S ratio, candidate / baseline (paired by instance; 16 targets × 3 rounds; 95% interval; green = below 1)

curve	pair	w = 0	w = 0.1	w = 1
p36	res1dp/res1	0.880 [0.752, 1.030]	0.425 [0.363, 0.497]	0.132 [0.112, 0.154]
p36	res1eq/res1	0.639 [0.561, 0.727]	0.346 [0.304, 0.394]	0.158 [0.138, 0.180]
p36	res2dp/res2	0.838 [0.777, 0.904]	0.346 [0.321, 0.373]	0.099 [0.092, 0.107]
p36	res2eq/res2	0.654 [0.588, 0.728]	0.303 [0.272, 0.337]	0.127 [0.114, 0.141]
p36	res4dp/res4	0.871 [0.786, 0.964]	0.279 [0.252, 0.308]	0.073 [0.066, 0.081]
p36	res4eq/res4	0.679 [0.612, 0.752]	0.243 [0.219, 0.270]	0.092 [0.083, 0.102]
p37	res1dp/res1	0.816 [0.679, 0.980]	0.392 [0.327, 0.471]	0.119 [0.099, 0.143]
p37	res1eq/res1	0.660 [0.561, 0.776]	0.358 [0.304, 0.421]	0.163 [0.138, 0.192]
p37	res2dp/res2	0.941 [0.821, 1.079]	0.386 [0.337, 0.443]	0.108 [0.094, 0.124]
p37	res2eq/res2	0.671 [0.562, 0.802]	0.310 [0.259, 0.371]	0.129 [0.108, 0.155]
p37	res4dp/res4	0.895 [0.800, 1.002]	0.284 [0.254, 0.318]	0.072 [0.064, 0.081]
p37	res4eq/res4	0.660 [0.590, 0.737]	0.235 [0.211, 0.263]	0.088 [0.079, 0.099]

Frozen rule C1 (all 24 cells below 1 at w = 0, 0.1, 1): **not met**. All 12 eq cells hold. 9 of 12 dp cells hold; the three failures are all at w = 0 (p36 res1dp upper bound 1.030, p37 res2dp 1.079, p37 res4dp 1.002). A/A (res2 vs res2aa) = 1.0000 on both curves; 1408/1408 records verified; `ecbench verify --replay 12` audits OK.



Distance remaining to matched rho (S ratio rho → arm; 1.00 = parity)

curve	arm	w = 0	w = 0.1	w = 1
p36	res4	3.97	12.62	51.57
p36	res2	3.12	7.65	28.04
p36	res1	2.42	5.04	16.87
p36	res4dp	3.46	3.52	3.77
p36	res2dp	2.62	2.64	2.77
p36	res1dp	2.13	2.14	2.22
p36	res4eq	2.70	3.07	4.76
p36	res2eq	2.04	2.32	3.56
p36	res1eq	1.54	1.75	2.66
p37	res4	4.00	12.71	51.92
p37	res2	2.90	7.11	26.07
p37	res1	2.49	5.19	17.37
p37	res4dp	3.58	3.61	3.74
p37	res2dp	2.73	2.75	2.81
p37	res1dp	2.03	2.04	2.07
p37	res4eq	2.64	2.99	4.58
p37	res2eq	1.95	2.21	3.36
p37	res1eq	1.64	1.86	2.83

Rho is ahead of every arm at every weight. The best arm at w = 0 is res1eq (1.54× p36, 1.64× p37 rho) and at w = 1 res1dp (2.22×, 2.07×); baseline res4 was 3.97 / 12.6 / 51.6× (p36) and 4.00 / 12.7 / 51.9× (p37).

eq versus dp, same B

curve	pair	w = 0	w = 0.1	w = 1	order
p36	res1eq/res1dp	0.726 [0.642, 0.820]	0.815 [0.721, 0.921]	1.201 [1.061, 1.359]	unresolved inside the band

p36	res2eq/res2dp	0.780 [0.699, 0.871]	0.876 [0.784, 0.978]	1.282 [1.145, 1.435]	unresolved inside the band
p36	res4eq/res4dp	0.780 [0.695, 0.875]	0.874 [0.778, 0.981]	1.262 [1.123, 1.418]	unresolved inside the band
p37	res1eq/res1dp	0.809 [0.699, 0.935]	0.911 [0.788, 1.054]	1.364 [1.177, 1.580]	unresolved inside the band
p37	res2eq/res2dp	0.713 [0.613, 0.830]	0.803 [0.690, 0.935]	1.196 [1.025, 1.395]	unresolved inside the band
p37	res4eq/res4dp	0.737 [0.669, 0.811]	0.829 [0.752, 0.913]	1.225 [1.111, 1.350]	unresolved inside the band

eq has the lower count at w = 0 and dp at w = 1 (eq pays about 2.5 priced operations per step against 1); the order is unresolved inside the band, so neither is ranked above the other.

Classification and caveats

- **Engineering** improvement of a generic collision search: the collector is a rho walk that needs B + 1 collisions, so its floor is about √(B + 1) times rho. At B = 1 the arms sit at 1.5–2.2× rho. It is not a structural index-calculus advance and does not beat rho.
- Per-step pricing is deliberately conservative: a DP test, an orientation read and negations are each charged 1 GAE at w = 1, where rho's tuned walk charges about 1 operation per step in total. A cheaper model (DP as a bit test on the canonical x, as in rho) would lower the eq w = 1 figures; it was not applied here.
- Only p36 and p37 (both ≈ 2^36) exercise the DP path; larger r is unmeasured. Local L0 run, counts only. No `VALIDATION.json`, so no leaderboard or `benchmarks/README.md` row.
- Exploratory, before this study: a first equivariant attempt with 16 jumps fell into short cycles (walks trapped until the cap) and lost; the 64-jump pair-sum step set with a 2-cycle escape fixed it.

Sources

Teske, Math. Comp. 70 (2001) 809–825; Bos–Kleinjung–Lenstra, ANTS-IX, LNCS 6197 (2010) 66–82 (fruitless cycles of the negation map; checked by search earlier in this session); van Oorschot–Wiener, J. Cryptology 12 (1999) 1–28 (distinguished points; cited from memory). Earlier work: `research/ic_residual_fused_20261031`, `research/ic_residual_dp_20261101`, `research/ic_prime_residual_lp_20261026`.